

As a consultant, I would say this application does a great job at using machine learning to predict the number of bottles sold in for a given month. However, it is important to highlight the data set the model was trained with. This dataset is part of Google's Big Query and is called Iowa Liquor Sales. It includes information on the name, kind, price, location of sale, and quantity of sales of individual containers and packages of containers of alcoholic beverages. This application focuses specifically on the number of bottles sold in a given month for all liquor stores across the state of Iowa. The training dataset is well explained, and it is also demonstrated on how it was acquired. It included total bottle sales each month from January 2015 to January 2025. The query to get the data is already performed and the JSON output is included. This is a nice advantage of the application because it allows us to go query the data on our own or just use the training data provided. Another big advantage of this application is that it includes containerized deployment. It is packaged using Docker, allowing it to run consistently across different environments. This application also has built-in monitoring integration with Prometheus. It enables real-time monitoring of application performance and system metrics. This helps identify performance bottlenecks and system health issues early. One last advantage is the implementation of automated continuous integration. The project includes a CI pipeline using GitHub Actions, which automatically builds and tests the application when changes are pushed. This improves reliability and reduces the risk of introducing errors into the codebase. Moving on to some of the disadvantages of this application, one of the more notable ones is the limited variables that were worked with. The Iowa Liquor Sales dataset has a lot more variables available to work with and this application only focuses on Month,

Year, and bottles sold. This leads to a weaker model due to the limited variables used. Also, by using more variables the model can be used to predict more than just bottles sold in a given month. This would be the strongest area of improvement. I would strongly recommend the training data be expanded to include more variables offered from the Big Query dataset. Using this updated training data the ML model can be improved as well. There would only need to be slight adjustments to the other features of the application such as the containerized deployment, built-in monitoring, and automated continuous integration. One other disadvantage of this application is the absence of a full continuous deployment pipeline. While the project includes continuous integration through GitHub Actions to automatically build and test the application, deployment of new versions still requires manual steps. A potential area of improvement would be implementing a continuous deployment workflow that automatically deploys updated container images to a platform such as Kubernetes, enabling automated releases and improved scalability. Another technology to be used in the future would be deploying the application on a managed container platform such as Google Cloud Run. Cloud Run allows developers to deploy containerized applications without needing to manage the underlying servers or infrastructure. Overall, this application is a great starting point for building a more advanced ML model on the Iowa Liquor sales dataset. The current application is demonstrated to work, and the successful operation is displayed. There is a clear prediction made from a model that is trained on part of a historical data set. There are also a handful of technologies that are successfully used and demonstrated in this application.

As the client you can use this application to accurately predict the number of bottles that will be sold in a specific month across the state of Iowa.